

CELLGUARD · REQUIREMENTS

CellGuard requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/cellguard		
Source	docs/03-requirements.md @ 73d539d		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-25	Amish Chadha	Initial scaffold

CellGuard requirements

These are first-pass requirements for the concept. Targets are proposals for review, not user-validated needs. They will be checked by calculation at TRL 3 and revised after the lab projects that depend on CellGuard confirm their needs. Status in Table 2 is judged against the estimates in CGD-PRC-001.

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)
R1	Cover the lab's LFP packs with one board	4 to 16 LFP cells in series (12.8 to 51.2 V nominal, 58.4 V maximum), cell count set in a configuration file	Design review against the front-end data sheet
R2	Measure every cell voltage accurately	± 10 mV at 25 °C and ± 15 mV from –20 to 60 °C, each cell at least every 100 ms	Data sheet review; later comparison with a calibrated meter
R3	Protect the pack from abuse	Cell overvoltage (default 3.65 V), cell undervoltage (default 2.50 V), charge overcurrent, discharge overcurrent, short-circuit trip within 500 μ s, over-temperature, charge lockout below 0 °C; all thresholds in the configuration file	Protection table review; later fault injection on a cell simulator
R4	Stay safe after a single fault	No single failure (microcontroller crash, stuck-on MOSFET, open sense wire, failed sensor) leaves the pack able to be overcharged or over-discharged	Failure mode and effects analysis
R5	Carry the lab's load currents	40 A continuous and 80 A for 10 s at 40 °C ambient, MOSFET junction at 110 °C or less	Thermal calculation; later bench test
R6	Waste little power in the power path	4 W or less on the board at 40 A (MOSFETs, shunt and copper), excluding the external fuse	Loss calculation from data sheets
R7	Balance cells	Passive balancing at 100 mA or more per cell; corrects a 1 % capacity imbalance in 24 h of balancing time on packs up to 100 Ah; board stays at 70 °C or less while balancing	Balance power and time calculation
R8	Give a state of charge the user can trust	Documented method; error within ± 5 percentage points after a full charge, within ± 10 points after 7 days of partial cycling without a full charge	Algorithm note; later cycling test against a lab reference
R9	Measure current	± 1 % of reading ± 50 mA from –80 to +80 A	Shunt and front-end error budget
R10	Draw little from an idle pack	300 μ A or less in sleep with outputs on and CAN able to wake; 10 μ A or less in ship mode	Quiescent current budget from data sheets
R11	Talk to host devices openly	CAN 2.0B at 250 or 500 kbit/s with a published message set; 3.3 V UART at 115,200 baud for logs and configuration; an enable input that keeps the output off until the host is connected	Interface specification review
R12	Keep a state-of-health log	At least 2,000 charge and discharge event records in non-volatile memory, exportable as CSV; fields sufficient for EU Batteries Regulation state-of-health data	Log format review
R13	Precharge the load	Charge up to 2 mF of load capacitance to 90 % of pack voltage within 1 s before closing the discharge path	Precharge calculation
R14	Be small and light	230 x 120 x 40 mm or smaller, 0.6 kg or less without power cables	Massing model; later weighing
R15	Be buildable and open	No BGA or leadless parts finer than 0.5 mm pitch; assembly with a hot plate or hot-air station; hardware CERN-OHL-S-2.0, firmware MIT	Design review of the parts list

ID	Requirement	Target	Verification (TRL 3 or later)
R16	Stay within the concept budget	\$120 or less in parts at quantity one, excluding cells	Priced BOM

Table 2. Status at TRL 2 (estimates).

ID	Status	Basis
R1	Met by design	Front end rated for 3 to 16 series cells
R2	At risk	The front end's data sheet gives better than 10 mV as a typical figure; the worst case over temperature is not yet checked
R3	Met by design	Front-end protections plus firmware thresholds; trip times to be confirmed at TRL 3
R4	Not met	The front end protects independently of the microcontroller, but a shorted charge MOSFET or a failed front end leaves no second layer except the charger's own voltage limit. Closing this needs an independent secondary protector (proposed, awaiting Amish)
R5	Met on estimate, unverified	About 2.0 W in the MOSFETs at 40 A at 25 °C, about 3 W when hot; heat spread into the base plate
R6	Met on estimate, thin margin	About 3.2 W at 40 A (about 4.2 W with hot MOSFETs, which would miss R6)
R7	Met up to 100 Ah; not met for 280 Ah cells	1 Ah at 100 mA takes 10 h; 2.8 Ah on a 280 Ah cell takes 28 h
R8	At risk	Depends on coulomb counter offset and how often the pack reaches full charge; unverified
R9	Unverified	Error budget not yet done
R10	Met on estimate	About 100 µA in sleep (estimate from typical data sheet figures)
R11 to R13	Met by design	See CGD-PRC-001
R14	Met	Massing model about 220 x 110 x 32 mm, about 0.5 kg (estimate)
R15	Met by design	Front end and microcontroller in 0.5 mm pitch quad flat packages
R16	Not met	About \$124, about 3 % over budget; about \$134 with a secondary protector

■ Assumptions

- LFP cell limits are typical defaults (3.65 V maximum and 2.50 V minimum per cell; charge 0 to 45 °C; discharge –20 to 60 °C). Every build must take its limits from the actual cell data sheet.
- Reference packs: 4S 280 Ah (12.8 V, about 3.6 kWh) for storage and 16S 20 Ah (51.2 V, about 1 kWh) for vehicles.

- The 40 A rating covers a 400 W inverter on a 12.8 V pack (about 31 A) and a 30 A vehicle controller. Larger loads need a larger board or a contactor, which is out of scope.
- Prices are indicative single-unit prices from distributors in September 2026 and will change.